

B2 BACnet / Modbus

Binary numbers

March 4th – 5th, 2026

Binary numbers

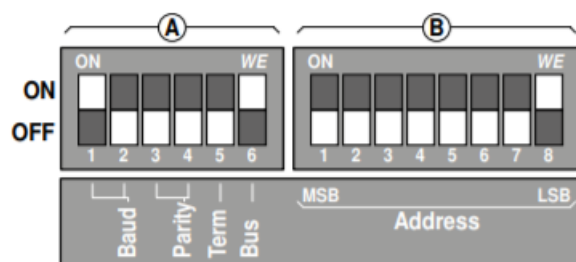


<https://www.youtube.com/watch?v=LpuPe81bc2w&t=252s> (up to 1'50'')

<https://www.youtube.com/watch?v=6WsI95N0QKU&list=PLqYRP-hBYN3SR8WSfbuWJRwwwVilbRgSO> (German language)

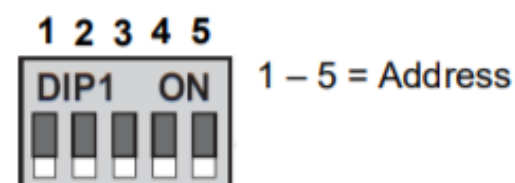
Binary numbers in use

Device addressing (i.e., Sensors, BKN)



BACnet address	1	2	3	4	5	6	7	8
0		OFF	OFF	OFF	OFF	OFF	OFF	OFF
1		OFF	OFF	OFF	OFF	OFF	OFF	ON
2		OFF	OFF	OFF	OFF	OFF	ON	OFF
...								
127		ON	ON	ON	ON	ON	ON	ON

Modbus address	1	2	3	4	5	6	7	8
1	OFF	OFF	OFF	OFF	OFF	OFF	OFF	ON
2	OFF	OFF	OFF	OFF	OFF	OFF	ON	OFF
3	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON
...								
247	ON	ON	ON	ON	OFF	ON	ON	ON



2^0 (1)	2^1 (2)	2^2 (4)	2^3 (8)	2^4 (16)	Address
1	2	3	4	5	
OFF	OFF	OFF	OFF	OFF	0
ON	OFF	OFF	OFF	OFF	1
OFF	ON	OFF	OFF	OFF	2
ON	ON	OFF	OFF	OFF	3
OFF	OFF	ON	OFF	OFF	4
ON	OFF	ON	OFF	OFF	5
OFF	ON	ON	OFF	OFF	6
⋮	⋮	⋮	⋮	⋮	⋮
ON	ON	ON	ON	ON	31

Binary numbers in use

Modbus EV – Malfunction and Service Info

105	104	Malfunction and Service Information Value is bit-coded. More than one bit can be set to 1 All bits not mentinoed in the enumeration are not used for this actuator range Error Sensor T1: Error with remote temperature sensor Error Sensor T2: Error with embedded temperature sensor Error Flow Sensor: Error with the flow sensor Actuator can't move: Mechanical overload due to blocked valve, etc. Flow with closed valve: Flow is measured but position of valve is closed Airbubbles: Air bubbles in the hydronic system. As long as there are airbubbles in the system, position control mode is active, regardless off control mode setting (Register No. 141). Flow not reached: Setpoint cannot be reached within 5 min during flow control Power not realized: Setpoint cannot be reached within 5 min during power control Gear disengagement active: Gear disengaged button is pressed Reverse flow: Reverse flow is detected MP communication faulty: Internal communication between sensor and actuator faulty Freeze warning: Measured temperature & glycol concentration indicate that grease ice can build up	Bit0: Error Sensor T1 Bit1: Error Sensor T2 Bit2: Error Flow Sensor Bit3: Actuator cannot move Bit4: Flow with closed valve Bit5: Air bubbles Bit6: Flow not reached Bit7: Power not realized Bit9: Gear disengagement active Bit11: Reverse flow Bit12: MP communication faulty Bit13: Freeze warning
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Binary numbers in use

IP Subnet mask

Subnet mask 255.255.255.0

IP-Address 192.168.178.125



Subnet mask: The first 3 number blocks

Network ID: 192.168.178.

Host ID: 125

The network includes: 192.168.178.1 – 192.168.178.254

Calculator

